

Flexible Work Arrangements and Firm Outcomes: Experimental Evidence from India

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July 18, 2025

Abstract

This paper studies the impact of flexible work arrangements on worker and firm productivity in the context of a large apparel manufacturing cluster in India. In the status quo, production line workers (tailors) are hired on piece rate contracts, implying that (1) workers are free to choose their work schedules, resulting in significant dispersion in worker arrival times at the firm, and (2) workers are only incentivized on individual output, as opposed to firm output. I implement an experiment with 60 small factories and 370 workers, where I incentivize all workers in treatment firms to arrive at work by the owners' preferred start time for a period of 4-weeks. I collect detailed data on daily attendance, work timings and production at the worker and firm level, starting 2 weeks prior to the introduction of worker incentives. This allows me to estimate the effect of the incentives on (a) worker labor supply, output and earnings, and (b) firm-level output and revenues. I find that incentives increase the likelihood that workers arrive on or before the preferred start time in treatment firms by 25% relative to control firms. These changes are driven by workers who were consistently late to arrive at the firm during the pre-incentive weeks (on whom the treatment was most binding), and female workers (who were more likely to be late at baseline). There is no effect on worker attendance, daily work hours and the total duration of breaks during the day. These changes in workers' labor supply add up to a significant improvement in worker coordination at the firm level: dispersion in worker arrival times decreases by 15-36% based on several measures of dispersion. I find a statistically significant increase in both weekly firm and worker output by 34% and 29% respectively. Weekly worker earnings increase by 23%, and weekly firm revenues increase by 30%. Unlike the effects on timely arrival, treatment effects on worker output and earnings are experienced by all workers in the firm, not just by late or female workers on whom the treatment was binding. This suggests two possible mechanisms behind these effects: (1) treatment changed the task allocation among workers, (2) treatment created positive spillovers to downstream workers in the production line, or both.

*shreya.chandra@berkeley.edu, PhD Candidate, Agricultural and Resource Economics, University of California, Berkeley. I am extremely grateful to my advisors, Jeremy Magruder, Aprajit Mahajan, Supreet Kaur and Ted Miguel for their advice and encouragement throughout this project. I would also like to thank Sydnee Caldwell, Marco Gonzalez-Navarro, Ethan Ligon, Alex Mas, Utkarsh Kumar, Benjamin Roth, Natalia Rigol, Pierre Biscaye, Shreya Sarkar, Nick Swanson, David Wu, and participants at the ARE Development Workshop, Berkeley Development Lunch, and EUDN PhD Workshop for many helpful comments and discussions. I am extremely grateful to K. Prakash, Apoorva Limaye, Sanskruthi Kalyankar, Nitesh Bharadwaj and Jayashree Devaraj for their time, support and hard work on this project. This research is funded by the Weiss Fund for Research in Development Economics, International Growth Center, Private Enterprise Development in Low-Income Countries and the Center for Effective Global Action at UC Berkeley. IRB approval was obtained from the Committee for Protection of Human Subjects at UC Berkeley and Institute for Financial Management and Research, India. AEA Trial Registration # AEARCTR-0015588.

1 Introduction

As countries develop, they undergo structural changes in the organization of labor – in particular, a shift from self-employment to employment in firms (Bandiera et al., 2022). Understanding frictions to wage-employment is a crucial topic of research in developing countries. Recent work attributes these low rates of wage-employment to workers’ preferences for flexibility (Cefala et al., 2024; Oh et al., 2024). While flexibility is an important job attribute for workers - especially so in low-income contexts, where prevalence of health shocks, lack of formal social protection and care infrastructure, reliance on kinship networks may reinforce such preferences (Breza and Kaur, 2025), flexible work arrangements may be costly or infeasible for firms. This is especially true in settings where the production process requires a high degree of coordination among workers (assembly line manufacturing), and may have broader implications for firm growth if coordination costs are increasing in scale (Lucas, 1978).

This paper studies the trade-off between firm and worker preferences, shedding light on a new type of labor market friction that affects firm productivity. I study the context of a large apparel manufacturing cluster in Tirupur, India which features many small-medium firms employing 5-15 workers (tailors).¹ Firms are organized as batch processing lines, where each worker (tailor) performs a distinct operation on the garment in batches of 80-100 pieces. The status quo work arrangement in this setting reflects the trade-off between firm and worker preferences. As is common in many developing country settings (Bassi et al., 2023), tailors are hired on piece rate contracts and organize as independent contractors in firms. This enables them to choose their work timings, instead of being required to follow uniform work shifts like their counterparts working in large/export firms.² Effectively, this means that workers do not coordinate their work schedules – in data from 60 firms and 370 workers, the difference in arrival times from company’s stated start time ranges from 60 minutes early to 75 minutes late (90-10 gap, see Figure 1). Moreover, because workers are only incentivized on individual output as opposed to team output (e.g. only 100 sleeves + 100 collars versus 100 pieces of the completed garment), workers’ individual choice of work schedules may create a coordination externality on both other workers and the firm owner.

On the other hand, the production technology implies that worker coordination may improve production - because operations need to be performed sequentially to produce the final output, the lack of coordination in worker’s arrival times can either (a) limit worker’s ability to specialize on 1

¹These firms account for $\approx 30\%$ of all 2+ worker firms in this district and 3-digit industry group (Economic Census, 2013)

²In another ongoing project in Tirupur, I document large and systematic differences in work arrangements by firm size – see Table A6. Interviews with both small and large firm owners suggests that differences in the provision of flexibility arise from a combination of institutional and technological constraints: (a) labor laws and social compliance from international buyers require large/export firms to systematically record work shifts in order to provide workers government benefits and overtime pay, and (b) large/export firms are more likely to organize production in assembly lines, while small firms follow batch production processes. Although both forms of production involve a high degree of task specialization among workers (i.e. workers perform only 1 distinct task per product/day), assembly lines have a stronger requirement for worker coordination than batch production processes.

task because they need to fill in for a late worker who usually performs the preceding task, or (b) increase idle time if they need to wait for the preceding task to be completed. Thus, such flexible work arrangements may slow the overall speed of production, and lower productivity by limiting specialization. This is consistent with owner’s beliefs about how flexibility affects production, and why imposing fixed work schedules may increase production: at baseline, owners were asked what type of measures related to their workers would help increase production (un-prompted). 90% of owners reported that they would like their workers to start work at the same time. This far outweighs other measures – 38% of owners would like workers to take fewer/shorter breaks, 25% would like workers to take fewer un-announced leaves and 17% to not bring their children to work (Figure 2a). When asked why coordinated timings would help increase production, 77% of owners chose idle time, and 67% chose task specialization as possible mechanisms (Figure 2b).

I conduct an experiment with 60 firms and 370 workers to test whether flexibility in work schedules affects firm productivity. I incentivize workers in treatment firms to follow fixed work start times for a period of 4-weeks. Treatment incentives are designed as a daily bonus of Rs. 100 (15% of average daily earnings), paid conditional on arriving at the firm by the owners’ pre-specified work start time, elicited at the time of baseline. Thus, the incentive encourages both attendance and before-time arrival. Workers in control firms receive unconditional incentives during the treatment incentive period, matched to a randomly chosen treatment firm each week to address concerns about income effects on worker productivity.

I collect daily data on worker attendance, arrival times, breaks taken, and production starting 2-weeks before the beginning of incentives through the 4-week incentive period. Compliance with the incentive is defined as being present, *and* arriving on or before the preferred start time. This allows me to estimate how costly it is for firms to incentivize their workers to arrive on time - in other words, an estimate of workers’ willingness to pay for flexibility. I use data on individual worker output to construct a measure of overall firm output - this allows me to test whether the incentives affect firm output.

I find that treatment significantly increases the probability that workers arrive at work at or before owners’ preferred start times as elicited at baseline. This effect is large - the fraction of worker days that workers are present and arrive before the preferred time increases by 11 percentage points, a 25% increase relative to a control mean of 45% before the incentives were active. I find no effect on worker attendance, suggesting that the effect on worker compliance is only driven by changes in workers’ arrival times. Heterogeneity analysis suggests that the effect on compliance is almost entirely driven by workers who consistently arrived after the preferred start time at baseline (hereafter, “late” workers), and female workers, who are more likely than male workers to be late at baseline. Workers in treatment firms do not reduce total daily work hours or increase the number and total duration of breaks to compensate for arriving early. Overall, these changes in worker compliance reduce dispersion in worker arrival times at the firm level - total dispersion in worker arrival times, measured as the difference in arrival times between the earliest and the last

worker, decreases by 25 minutes, or 20% relative to the control mean (this measure is analogous to the spread of the distribution in [Figure 1](#)). The deviation from preferred start time for the 90th percentile worker - which indicates the 2nd to last worker to arrive given firms have an average of 6 tailors - decreases by 36% relative to the control mean. Thus, incentives are successful in improving coordination in worker arrival times in treatment firms.

Next, I estimate the intent-to-treat effects of the incentive on worker output and earnings. I find no statistically significant increase in daily worker output in treatment firms. However, weekly worker output in treatment firms increases by 0.28 standard deviations (30% relative to the control mean, equivalent to 1,000 more pieces per week). Surprisingly, unlike the results on labor supply, increases in weekly worker output and earnings are realized by all workers, and not driven by female or late workers. This suggests

Finally, I estimate the effect of incentives on overall firm output, measured as the total number of finished pieces produced by all workers in the firm (worker output is the sum of total pieces produced by a worker, across all tasks/products, regardless of whether the garment has been fully stitched). Consistent with the increase in weekly worker output, I find a statistically significant, 0.31 standard deviation increase in weekly firm output – 34% increase relative to the control mean of 3,200 and control standard deviation of 3,500, equivalent to an increase in total weekly output by 1,000 pieces. Weekly firm revenues increase by 0.21 standard deviations (control mean of INR 52,000, control SD of INR 76,000), and is marginally significant (p-value of 0.10). This translates to a 30% increase in firm revenues in treatment firms during the incentive period.

This paper makes contributions to 3 strands of the literature: First, I experimentally estimate workers’ demand for flexible work arrangements. While this has been widely documented across many contexts ([Mas and Pallais, 2017](#); [Ho et al., 2024](#); [Baysan et al., 2024](#); [Wiswall and Zafar, 2018](#)), this paper’s contribution is to link worker’s WTP for flexibility with firm outcomes. This result will highlight an important, yet under-explored relationship between the provision of flexible work arrangements, and labor supply preferences ([Mas and Pallais, 2020](#)).

Second, I build on previous work documenting search and hiring frictions in low-income countries ([De Mel et al., 2019](#); [Hardy and McCasland, 2023](#); [Bassi and Nansamba, 2022](#); [Caria and Falco, 2024](#); [Carranza et al., 2020](#)) by documenting that even conditional on forming matches, firms may face barriers to coordinating workers. This could have wider economic consequences on firms’ ability to scale their workforce as coordination becomes costlier with scale ([Lucas, 1978](#)).

Finally, this project will contribute to our understanding of the internal organization of small, informal firms in developing countries, and the role of underlying market frictions, building on recent work such as [Bassi et al. \(2022, 2023\)](#). Specifically, this project will document worker preferences as the source of coordination costs for SMEs in developing countries, and how they affect firms’ ability to specialize.

The remainder of the paper is organized as follows: [section 2](#) describes the study context and sample firms, [section 3](#) describes the experimental design, implementation and data collection, and [section 4](#) lays out the empirical strategy. [section 5](#) presents the main results.

2 Study Context and Sample

The context of my study is Tiruppur district in the state of Tamil Nadu, India. This region is part of a major garment manufacturing hub in India, catering to both export and domestic markets since the 1970s. The production process in Tiruppur is highly decentralized and carried out by sub-contracted units, called job-work units - these units will form my study sample. Job work units are *small* factories, employing about 10-30 workers, at least 50% of whom are women.³

Firms are organized as batch processing lines, where each worker performs a distinct operation on the garment in batches of 80-100 pieces. Operations need to be performed sequentially in order to produce a finish piece of garment. Firms typically specialize in broad product categories - such as t-shirts, bottomwear, innerwear, or kids wear. The average weekly order size in these firms is about 4,000 pieces, varying with product type.

The study takes place in 4 neighborhoods located in Tiruppur North sub-district, Tiruppur, Tamil Nadu. These neighborhoods were identified as high density clusters of small-medium job-work units during prior scoping work and area mapping exercises. I used the following criteria to screen firms to be part of the sample:

1. Has a stitching section
2. Employs at least 2 tailors who are not related to the firm owner or who work for pay (piece rate contract)
3. Must deliver finished products back to the buyer/retailer
4. Must receive cut-pieces of cloth to be stitched (ie does not only produce own designs)

All firms that satisfy these criteria were eligible for baseline. In case eligible firms were located adjacent to each other, I approached alternate firms in order to minimize spillovers between treatment and control firms located next to each other.

3 Experiment Design

Firms are randomly assigned to one of two treatment groups - (T) Treatment ($n = 30$), and (C) Control ($n = 30$). The treatment group was further randomized into (T1) high and (T2)

³Note that these firms are of course larger than the typical small developing country firm; however they employ a substantial fraction of the workforce within the textile industry in Tiruppur district - as of 2013, the share of total employment in textiles in 10-30 worker firms ranged between 20-30% in rural and urban areas in Tiruppur district (source: Economic Census, 2013-14). Such organization structures are common in many industrial clusters around the world – see [Nadvi and Schmitz \(1994\)](#); [Banerjee and Munshi \(2004\)](#); [Chaudhry and Woodruff \(2013\)](#); [Atkin et al. \(2017\)](#).

low incentive sub-treatments, with daily bonus amounts set at Rs. 100 (high) and Rs. 50 (low) respectively. These amounts are approximately 15% and 7.5% of worker’s average daily earnings and were chosen based on owners’ estimates of the minimum amount needed to encourage workers to come to work early. Workers’ average daily earnings are Rs. 600. Treatment incentives are conditional on the worker arriving at the firm and starting work by the company’s preferred start time. Thus, the incentive encourages both on-time arrival and attendance. The total incentive amount was paid to workers as part of their weekly earnings.⁴ The unconditional incentive amount for control group firms was matched to the total realized payment made to a randomly selected treatment firm, capped at Rs. 300 per week (the maximum incentive amount in the low incentive treatment group T2). These treatment-control firm (randomized) pairings were created separately each week. [Figure 4](#) shows the experiment design.

3.1 Data and Summary Statistics

Baseline: I conduct a baseline survey with owners to capture owner and firm characteristics, and a complete worker roster, which is followed by a baseline survey of all workers at the firm. Randomization and treatment assignment was done after owner and worker baselines.

Baseline firm and worker characteristics, and balance by treatment arms are reported in [Table 1](#) and [Table 2](#) respectively. On average, firms hire 6 tailors (excluding owners/co-owners), and have been operating for 5-7 years. Firms produce 4,120 pieces of output each week, and earned a profit of INR 43,000 in the last 30 days (USD 490). There are some notable differences between control and treatment firms at baseline – for instance, treatment firms less likely to produce their own order, and export orders, and earn INR 2,300 less in revenues per worker (INR 1300 less in profits per worker). However, all differences, except the probability of producing own designs are statistically insignificant.

Owners directly supervise workers (that is, without a manager/contrator) in almost all firms – owners also perform stitching work in 90% of firms, and work as regular tailors in about half of firms. About 76% owners are male, 52% have completed secondary school and have 19 years of experience in the garment industry.

Workers are on average 37 years old (similar to owners), 44% are male and most workers are married (93%) and have a child younger than 14 years of age (86%). 84% workers live within a 15 minute commute to the firm, have worked at the current firm for 3 years and have 15 years of experience in the garment industry. Worker characteristics are balanced across treatment and control firms ([Table 2](#)).

Order, Task and Piece Rates Data: At baseline, I also record a roster of all styles that the firm is currently producing. For each style, I record the piece rate that the owner receives for a

⁴All workers who were working full-time at the firm at any point during the study were eligible to receive incentives. Workers’ full-time status was determined by the owner and verified by the research team using daily attendance data.

finished stitched piece and its task structure – a list of all tasks which they need to be performed to produce the finished product, the order in which they are performed, and their piece rates. This roster is updated throughout the 6-week study period to incorporate new styles that the firm is producing.

Daily Production and Attendance Data: Starting 1-2 weeks after baseline, I collect data *daily* for a period of 6-weeks, out of which workers are paid incentives for 4-weeks.⁵ Each day, I record worker’s attendance, arrival time, break times during the day, and daily output (by product-task). Arrival times and attendance are recorded by surveyor observation and are not prone to mis-reporting in order to game the incentive; remaining data is self-reported by workers. Daily output is recorded for all production workers, which includes the owner and/or co-owner in some firms. I use daily output data from all production workers to construct an independent measure of total firm production. In addition to tailoring/production output, I also record whether the worker did any other work at the firm, such as helper work, receiving or delivering raw material/finished pieces, machine repair, purchasing inputs, etc.

Table A1 summarizes key labor supply and production outcomes during the pre-incentive weeks, and differences across two worker types – gender, and whether the worker consistently arrived after the preferred start time during the pre-incentives weeks (hereafter, referred to as “late” versus “early” workers). Column (1) reports summary statistics for the overall worker sample; columns (2)-(3) report summary statistics for female and male workers, and columns (4)-(6) report summary statistics for early, “late”, and “very late” workers. “Late” and “very late” are indicators for workers whose average pre-incentive arrival times were between 0 and 30 minutes, and above 30 minutes of the preferred start time, respectively.

Female and male workers have similar demographics – the only exceptions being that women are more likely to be married, live closer to the firm, and have 5 years less experience in the industry than men. On the other hand, late and very late workers are more likely to be female, married, and have more children below 14 years of age. There are no differences in commute times between early, late and very late workers, suggesting that arrival times observed in status quo are not driven by commuting differences, but are likely driven by time commitments in the household before the start of the work day.

Pre-incentive attendance rates are similar across men and women, however, there are several notable gender differences in terms of arrival times and compliance: women arrived 21 minutes after the preferred start time on average, while men arrive 11 minutes before the start time – consequently, women were more likely than men to be “late” and “very late”, and 23 percentage points less likely to comply with the preferred start-time - one-third of men’s average compliance rate. On the other hand, attendance rates are the highest among late workers at 83%, 10 percentage

⁵In Phase 1, I collected data 1 week pre-incentives and 1 week post-incentives. In Phase 2, I collected 2 weeks of pre-incentives data only, dropping the post-incentive data collection.

points higher than attendance rates among early and very late workers.⁶ More surprisingly, only 65% of early workers, 45% of late workers, and 10% of very late workers complied with owners' preferred start times before the incentives – suggesting that (a) early workers are not the most reliable workers, and (b) late workers do comply almost half of the time. Given these patterns, I expect the incentive to encourage higher attendance rates among early workers, and to encourage both attendance and on-time arrival among late and very late workers. These patterns also suggest a possible trade-off between regular attendance and on-time arrival. Finally, I observe that women seem to compensate for their late arrival by taking fewer number of breaks during the day, so that average daily hours are only 40 minutes (0.6 hours) lower than men. Late workers work 40 minutes more than both early and very late workers, with no striking differences in the number or duration of breaks taken.

In terms of production outcomes, workers perform 1-2 tasks per day on average, and work on non-tailoring tasks 12% of the time. The average piece rate of tasks performed in a day is 80 paise, with some notable differences by gender – women work on tasks with 30 paise lower average piece rates than men; and baseline lateness – average piece rates decline monotonically with baseline lateness. All workers produce 100 pieces per hours, resulting in small daily earnings differences – women earn Rs. 150 less per day than men, and very late workers earn Rs. 200 less per day than early workers. These differences in piece rates and earnings partly stem from the type of tasks performed – men, and early workers are most likely to perform Complex Singer tasks before the incentives. These tasks are slower and have higher piece rates than the average task.⁷

Deliveries Data: At the end of each week, I record all orders received and delivered by the firm in the past week. I use information on total quantity delivered in a week as a separate measure of firm output and sales in that week.⁸ In addition to information on deliveries, owners are also asked to match each delivery to the date that material was received – this allows me to identify distinct “orders” received by the firm. Additionally, for all new materials received, I elicit owners' estimated planned delivery date, which I can compare with the actual delivery dates reported by owners to assess how treatment affects owners' planning of production.

Endline: Finally, endline surveys conducted during the last week of the study capture firm owner's and worker's perceptions of the treatment (in treatment firms), preferences for flexible

⁶This implies an overall average absenteeism rate of 24%, and 17% among late workers. This range is similar to absenteeism rates documented in other studies. Notably, absenteeism rates among late workers are similar to those reported in large, export factories – 11% in Tanzania (Ho and Huang, 2024), and 15% in India (Adhvaryu et al., 2022). The average absenteeism is similar to those observed in casual labor markets, which are closer to the labor market in this study (25% in rural casual labor markets in Burundi (Cefala et al., 2024), and 20-30% in urban casual labor markets in India (Cefala et al., 2024).

⁷About 50% of all singer tasks are complex tasks, and output per hour on Complex Singer tasks is half of that on other tasks.

⁸As discussed later in the results section, I use firm output constructed from production data as my preferred measure of firm output, since the quantity of deliveries may be affected by stock of finished pieces at the firm which I do not observe at the start of 6-weeks of daily data collection.

work arrangements, changes in firm operations during the study and time-use during and outside of work (relative to the pre-study period). These questions allow me to examine (1) the nature of workers' commitments/restraints that drive their preference for flexibility, (2) how compliance with the incentive impacted these commitments/restraints, and (3) whether compliance shifted workers' preferences for flexible (versus strict) work arrangements.

3.2 Implementation

The experiment was implemented in two phases - Phase 1 was implemented with 17 firms (9 treatment, 8 control) from November 2024 to January 2025. Phase 2 was implemented with 43 firms (22 treatment, 21 control) from March 2025 to May 2025. [Figure 5](#) shows the study timeline. Phase 1 only included high treatment firms (T1), and the half of the control firms did not receive any incentive (pure-control). In Phase 2, all control firms receive unconditional incentives matched to a randomly chosen treatment firm, up to the maximum allowable amount in low-treatment firms (Rs. 300). These changes were made to simplify field logistics by minimizing both distance between sample firms as well as firm attrition.

Randomization for Phase 1 was conducted by generating a random order of treatment and control assignments, which firms were matched to in the order in which their baseline survey was conducted. Randomization for Phase 2 was stratified on the degree of dispersion in worker arrival times around the owner's preferred work start time, using worker's usual arrival times as reported by the owner at baseline. All specifications reported in the paper control for phase by strata fixed effects to account for the method of randomization in each phase ([Bruhn and McKenzie, 2009](#)).

4 Empirical Strategy

I estimate the effect of incentives on worker level outcomes using the following specification:

$$Y_{i,f,t} = \alpha + \beta T_{i,f} \times \mathbb{I}\{\text{During}\}_t + \lambda_{i,f} + \mu_w + \mu_{dow} + \varepsilon_{i,f,t} \quad (1)$$

where $Y_{i,f,t}$ is the outcome of interest for worker i in firm f at time t , and time is either day or week, depending on the outcome variable. $T_{i,f}$ is an indicator that equals 1 if a firm f is assigned to treatment (fixed over time), $\mathbb{I}\{\text{During}\}_t$ is an indicator variable that takes the value 1 if the study week is an incentive week, and $\lambda_{i,f}$ denotes worker fixed effects. μ_w are study-week fixed effects,⁹ and μ_{dow} are day-of-week fixed effects (included only in day-level specifications). β is the coefficient of interest, and standard errors $\varepsilon_{i,f,t}$ are clustered at the firm level.

⁹Study weeks take the value 1 to 4 for the 4 incentive weeks, 0 for the week prior to incentives. Weeks 0-4 are common across Phase 1 and 2. The first pre-incentive week in Phase 2 is coded week -1, and the final, post-incentive week in Phase 1 is coded week 5. In the current analysis, I drop the post-incentives week (week 5).

I use a similar specification for firm-level outcomes:

$$Y_{f,t} = \alpha + \beta T_f \times \mathbb{I}\{\text{During}\}_t + \lambda_f + \mu_w + \mu_{dow} + \varepsilon_{f,t} \quad (2)$$

where indices and variables are defined as above, and standard errors $\varepsilon_{f,t}$ are clustered at the firm level. Some firm outcomes are only measured at endline, in which case I will run [Equation 2](#) without the t subscript and without firm and study-week fixed effects. Day-of-week fixed effects are only included in firm-day level regressions.

5 Results

5.1 Labor Supply

First, I test the effect of the incentives on worker attendance and compliance with work start times. Columns (1) and (2) in [Table 3](#) report results for worker attendance and compliance. Compliance is defined as an indicator for being present at the firm, *and* arriving by the owner’s preferred start time. I find no effect on attendance (column (1)), and an 11 percentage point increase in compliance (column (2)) in treatment firms during the incentives weeks. This represents a 25% increase in compliance relative to the pre-incentive control mean (45%), and is driven by workers arriving on time, rather than an a change in worker attendance. The effect on compliance is larger, and more meaningful when conditioned on workers being present at the firm – on average, 60% workers - or 3-4 out of 6 workers - arrived on time conditional on being present at the firm, and treatment incentives increase worker compliance to 76% (conditional on being present) – 4-5 out of 6 workers. That is, the treatment encourages almost all workers to arrive before the preferred start time (given the small size of the workforce).

Next, I examine whether these effects vary by gender, and baseline lateness in [Table 4](#). Treatment effects on attendance for women and “late” workers smaller than the respective base categories, however the differences are small in magnitude and insignificant. This is consistent with the result above that incentives did not change worker attendance in treatment firms, but it is surprising that attendance among early workers does not improve.¹⁰ On the other hand, compliance is driven entirely by female workers, and workers who were consistently late at baseline – female workers are 13 percentage points more likely than male workers to comply (column 3); “late” and “very late”

¹⁰37% of worker absences are because of social functions, followed by 19% because there was no work at the firm, 18% because either the worker or a family member was sick, and 10% because of local temple/religious festivals. A majority of these reasons relate to importance of social/kinship relations, as emphasized in [Breza and Kaur \(2025\)](#), but it is worth noting a large fraction of absenteeism attributable to low labor demand. This matches the rate of all-day firm closures recorded in the data - 10% of firms were closed for the entire day. 40% of closures were because the firm did not have work, followed by 36% because of local temple festivals or the firm owner’s personal commitments, and 10% because of power cuts.

workers are 17 and 35 percentage points respectively more likely than “early” workers to comply (column 4). This is expected, since the incentives were more binding on workers who consistently arrived after the preferred start-time, which was more common among women. In fact, this suggests that incentives may reduce the gap in work hours, and potentially earnings between men and women reported earlier (Table A1).

Do workers compensate for arriving earlier at the firm by working fewer hours, or taking more breaks during the work day? Table 5 reports treatment effects on daily work hours, the number and duration of breaks taken in a day. Daily work hours is defined as the total hours worked in a day, excluding the total time spent on breaks. Breaks include any time taken for lunch, tea, or other reasons for which the worker left the firm.¹¹ All outcomes are coded 0 on days that a worker was absent. Workers in treatment firms work 0.03 additional hours, or 1.8 additional minutes, relative to workers in control firms (column (1)), who work 6.7 hours per day on average pre-incentives. Similarly, I do not find any statistically or economically significant changes in the number of breaks (column (2)) or total duration of breaks (column (3)).

The treatment effects on work hours and break duration vary slightly across male and female workers, and “early” versus “late” workers; however, most of these differences are small in magnitude and statistically insignificant, with the exception of “very late” workers, who increase the total number of breaks taken in a day by 0.32 breaks (statistically significant at the 10% level). This is not surprising when looking at the treatment effect on deviation from the preferred start time, reported in Table A2. On average, workers in treatment firms arrived 8 minutes earlier relative to control workers, but this effect is driven by “late” and “very late” workers (column (3)). “Early” workers did not change their arrival times, while “late” and “very late” workers arrived 10 and 31 minutes earlier relative to control workers.

5.1.1 Does treatment improve worker coordination in arrival times?

Next, I examine how treatment affects worker coordination at the *firm-day level*, conditional on being present. Table A3 reports treatment effects for three measures of dispersion in worker arrival times at the firm-day level: minutes between the 1st and last worker to arrive at the firm (column (1)), the 90-10 gap in worker arrival times (relative to the preferred start time) (column (2)), and the deviation from preferred start time for the 90th percentile worker (columns (3)). This regression excludes firm-days on which only 1 worker was present at the firm, since the notion of coordination is not meaningful in such cases.

Total dispersion in worker arrival times is 124 minutes on average at baseline – treatment reduces dispersion by 25 minutes, a 20% decrease in dispersion, or improvement in coordination relative to control firms (column (1)). The 90-10 gap is a more conservative measure of dispersion which excludes extreme outliers – workers who arrive extremely early or extremely late. For example,

¹¹Workers rarely perform other work for the firm during breaks (3% of worker-days).

the 90-10 gap is 40 minutes lower than total dispersion in control firms during the pre-incentives week(s). Treatment further reduces the 90-10 gap by 13 minutes, a 15% decrease relative to the control mean of 82 minutes pre-incentives. However, this effect is not significant (column 2). Finally, I look at treatment effects on deviation from the owner’s preferred start time for the 90th percentile worker. This measure allows me to test whether these effects are driven by workers for whom the incentive was binding. Indeed, the deviation of the 90th percentile work reduces by 16 minutes in treatment firms, relative to the control group mean of 46 minutes pre-incentives. These effects amount to a 36% reduction in dispersion, or improvement in coordination as a result of treatment.

5.2 Worker Output and Earnings

Next, I test whether incentives increase worker output and earnings. Daily worker output is measured as the total quantity of pieces stitched by the worker in a day, across all tasks (and products). Note that this does not necessarily correspond to firm output, since a worker who stitched 100 collars and 100 sleeves would be considered to have worked on 200 pieces on that day. However, this measure aligns best with workers earnings, which are constructed as the product of the task specific piece rate and quantity produced on that task, summed over all tasks and products worked on in a day. I construct similar measures for weekly worker output and earnings. Output and earnings are coded as 0 on days that a worker was absent. Since firms produce different types of products, I normalize daily (weekly) worker output and earnings to pre-incentive, product level, daily (weekly) means and standard deviations. If a worker worked on multiple products in a day, I normalize by product and sum the normalized output across all products in a day.

Worker Output Results for daily and weekly worker output are reported in Panel A of [Table 7](#). The treatment effect on daily worker output is 0.08 standard deviations – relative to the control mean of 670 pieces, this amounts to a 9% increase in daily output, and is not statistically significant at conventional levels (column (1) of [Table 7A](#)). However, the effect on weekly output is much larger, and statistically significant at the 10% level – workers in treatment firms produce 0.28 standard deviations more per week, amounting to a 30% increase in weekly output relative to the control group mean (column (3) of [Table 7A](#)). The effects on daily and weekly output are not statistically different among females, or “late” and “very late ” workers compared to their male/early counterparts (columns (2)-(3) and (5)-(6) in [Table 7A](#)).¹²

Worker Earnings Results for daily and weekly worker earnings are reported in Panel B of [Table 7](#). The treatment effect on worker earnings mirrors the results on worker output – there is no statistically significant effect on daily earnings, which increase by 0.04 standard deviations (5% of the control mean, see column (1) of [Table 7B](#)). Weekly earnings increase by 0.21 standard deviations (23% of the control mean), but this effect is weakly significantly at the 10% level.

¹²These results look similar when I use actual levels of output by product type, and include product fixed effects (see [Table A4](#)).

Notably, treatment partly closes the weekly earnings gap between male and female and early and very late workers in pre-incentive weeks, as documented in [Table A1](#).

Surprisingly, and unlike the results on worker compliance, worker output and earnings are not only concentrated among either female or late and very late workers. This may be due to several reasons, which will determine further analysis and mechanism tests in the paper. Specifically, I hypothesize two reasons explaining the result on worker earnings: first, by changing worker arrival times (and consequently, both better coordination in arrival times, and a different order in which workers arrive), treatment may have resulted in a change in task allocation between workers in the firm. It is an open question whether the new task allocation is more efficient for workers and firms. I plan to investigate this further by exploring whether new task allocations are better aligned with worker’s expertise¹³, or increase the degree of task specialization in the firm. Second, the fact that workers for whom incentives were not binding also benefit from incentives suggests the presence of coordination externalities along the production. For example, increased compliance among late and/or female workers increases their own output each day, but may have spillover benefits to downstream workers (workers who perform the task next in the order). The firm may realize increases in production through the production line even without changes in the task allocation, or changes in the degree of task specialization. These results are work-in-progress and will be included in future revisions of the paper.

5.3 Total Firm Output and Revenue

How do the changes in individual worker output add up to total firm output and revenues? I construct two measures firm output: (1) using daily production data collected from workers, hereafter referred to as “Production Data”, and (2) using data on weekly deliveries reported by the firm owner, hereafter, “Deliveries Data”.

Production data is constructed by adding up total quantity produced on the last task in the production line, across all workers, for each distinct product. Total output is summed by broad product category over the week, and normalized to pre-incentive, product level, weekly means and standard deviations. Firms do not operate on all tasks of a product in a single day (owing to the small size of their workforce), so firm output is measured at a weekly level. Deliveries data is reported by the owner - during data collection, total deliveries and total production was cross-verified as part of regular data quality checks. However, deliveries data may still vary from production data if firms are delivering older inventory, or had inventory of finished products before I began collecting production data. I calculate weekly firm revenue as the product of total firm output and the piece rate that owners earn, and weekly labor cost as the product of total firm

¹³Every week, I record worker productivity at the task they are performing at that time. Productivity is measured as the number of pieces of that task completed within 5-minutes and is recorded by surveyor observation. I also elicited which tasks workers consider themselves to be an expert at in worker baseline surveys. I plan to use these data to assess whether treatment improves task allocation by making workers more likely to work on their expert tasks.

output and the total piece rate that workers earn (across the product’s constituent tasks) in each week.

Table 9 reports results on weekly firm output (columns (1)-(2)), revenue (columns (3)-(4)) and total labor cost (columns (5)-(6)). Odd (even) columns report outcomes constructed using the production (deliveries) data – I focus on outcomes constructed using the production data as my preferred measure of firm output since the quantity of deliveries may be driven pre-existing inventories that I do not observe in my data. To absorb week by week variation, I calculate average weekly production, revenues and labor costs before and during incentive weeks, and estimate a double-difference version of **Equation 2**.¹⁴

I find a 0.31 standard deviation increase in total weekly output in treatment firms during the incentive weeks – relative to the control mean and standard deviation, this is a 34% increase in total weekly firm output (control mean = 3,200). This effect is statistically significant at the 10% level, and is consistent with the effect on individual worker output. Weekly firm revenues increase by 0.21 standard deviations – 30% relative to the control group (control mean = INR 52,000 / USD 600), marginally significant with a p-value of 0.10; and total labor costs increase by 0.16 standard deviations – 21% relative to the control group (control mean = INR 19,000 / USD 220), not significant at conventional levels with a p-value of 0.28. These results provide suggestive evidence that improving worker coordination can increase firm output and profits, assuming that other fixed, non-labor costs did not change differentially between control and treatment firms. Further analysis will use self-reported profits, revenues, and non-labor costs from the owner endline survey to comprehensively assess the profitability of incentives, and compare these estimates to total weekly payouts made to workers to encourage them to arrive on time.

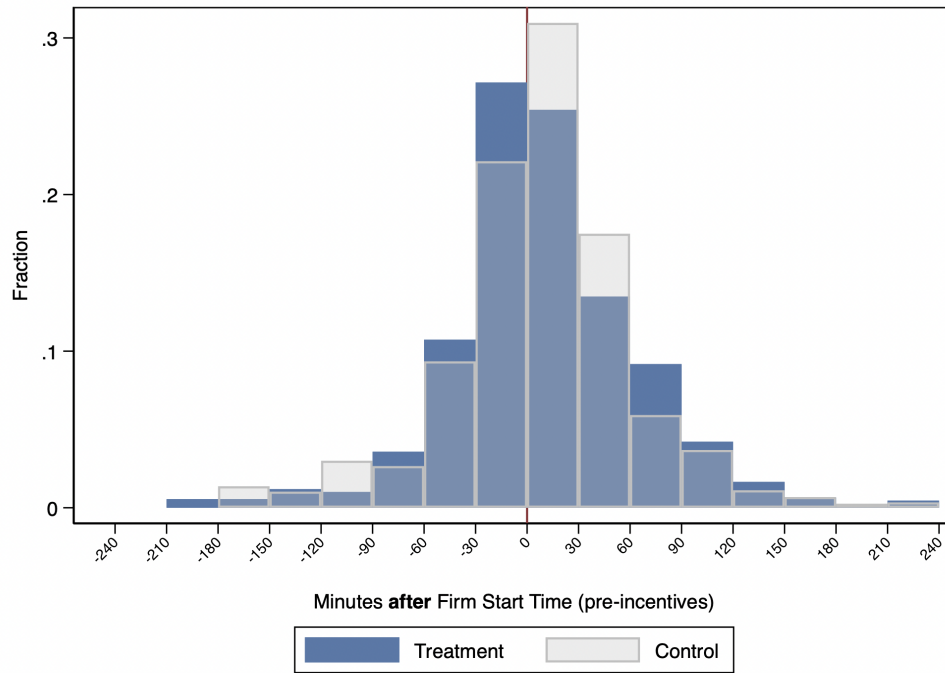
¹⁴Week-by-week estimates are reported in **Table A5**.

References

- Adhvaryu, A., N. Kala, and A. Nyshadham (2022). Management and shocks to worker productivity. *Journal of Political Economy* 130(1), 1–47.
- Atkin, D., A. Chaudhry, S. Chaudhry, , A. K. Khandelwal, and E. Verhoogen (2017). Organizational barriers to technology adoption: Evidence from soccer-ball producers in pakistan. *The Quarterly Journal of Economics* 1101, 1164.
- Bandiera, O., A. Elsayed, A. Heil, and A. Smurra (2022). Economic development and the organisation of labour. *Working Paper*.
- Banerjee, A. and K. Munshi (2004). How efficiently is capital allocated? evidence from the knitted garment industry in tirupur. *The Review of Economic Studies* 71(1), 19–42.
- Bassi, V., J. H. Lee, A. Peter, T. Porzio, R. Sen, and E. Tugume (2023). Self-employment within the firm. Technical report, National Bureau of Economic Research.
- Bassi, V., R. Muoio, T. Porzio, R. Sen, and E. Tugume (2022). Achieving scale collectively. *Econometrica* 90(6), 2937–2978.
- Bassi, V. and A. Nansamba (2022). Screening and signalling non-cognitive skills: experimental evidence from uganda. *The Economic Journal* 132(642), 471–511.
- Baysan, C., M. H. Dar, K. Emerick, Z. Li, and E. Sadoulet (2024). The agricultural wage gap within rural villages. *Journal of Development Economics* 168, 103270.
- Breza, E. and S. Kaur (2025). Labor markets in developing countries. *Annual Review of Economics* 17.
- Bruhn, M. and D. McKenzie (2009). In pursuit of balance: Randomization in practice in development field experiments. *American economic journal: applied economics* 1(4), 200–232.
- Caria, S. A. and P. Falco (2024). Skeptical employers: Experimental evidence on biased beliefs constraining firm growth. *Review of Economics and Statistics* 106(5), 1352–1368.
- Carranza, E., R. Garlick, K. Orkin, and N. Rankin (2020). Job search and matching with two-sided limited information.
- Cefala, L., S. Kaur, H. Schofield, and Y. Shamdasani (2024). Habit formation in labor supply.
- Cefala, L., P. Naso, M. Ndayikeza, and N. Swanson (2024). Under-training by employers in informal labor markets: Evidence from burundi. Technical report, Working Paper.
- Chaudhry, T. and C. Woodruff (2013). Incentives and productivity: Work groups vs. production lines. *IGC Working Paper (F-6005-PAK-1)*.

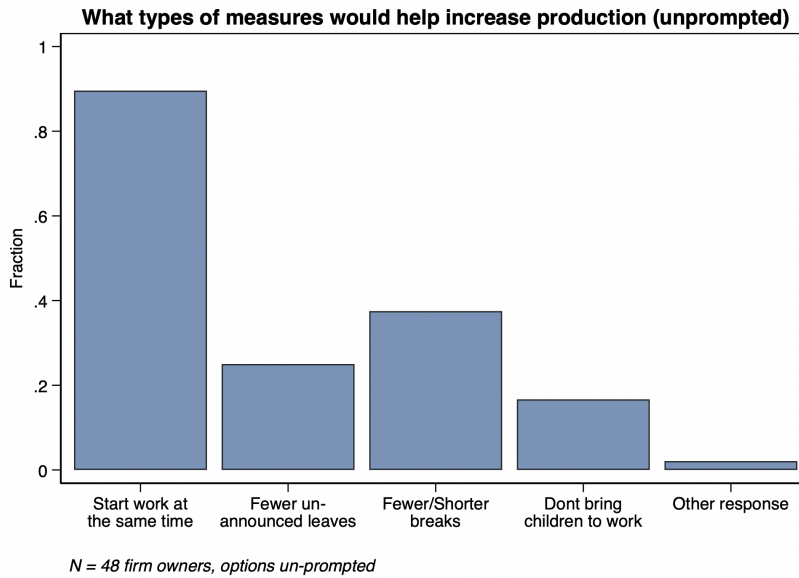
- De Mel, S., D. McKenzie, and C. Woodruff (2019). Labor drops: Experimental evidence on the return to additional labor in microenterprises. *American Economic Journal: Applied Economics* 11(1), 202–235.
- Hardy, M. and J. McCasland (2023). Are small firms labor constrained? experimental evidence from ghana. *American Economic Journal: Applied Economics* 15(2), 253–284.
- Ho, L., S. Jalota, and A. Karandikar (2024). Bringing work home: Flexible work arrangements as gateway jobs for women in west bengal. Technical report, Technical Report.
- Ho, Y. and Y. Huang (2024). Selection and sorting when supervisors have discretion: Experimental evidence from. *Working Paper*.
- Lucas, R. E. (1978). On the size distribution of business firms. *The Bell Journal of Economics*, 508–523.
- Mas, A. and A. Pallais (2017). Valuing alternative work arrangements. *American Economic Review* 107(12), 3722–3759.
- Mas, A. and A. Pallais (2020). Alternative work arrangements. *Annual Review of Economics* 12(1), 631–658.
- Nadvi, K. and H. Schmitz (1994). *Industrial clusters in less developed countries: review of experiences and research agenda*, Volume 39. Institute of Development Studies Brighton.
- Oh, S., Y. Shamdasani, and S. Goraya (2024, September). Worker absences and demand for flexible contracts.
- Wiswall, M. and B. Zafar (2018). Preference for the workplace, investment in human capital, and gender. *The Quarterly Journal of Economics* 133(1), 457–507.

Figure 1: Dispersion in Worker Arrival Times, before Incentives

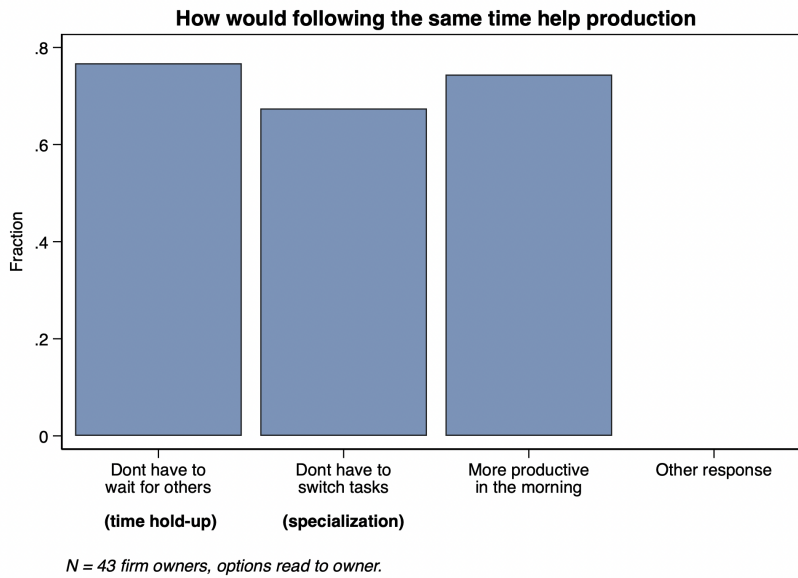


Notes: This figure plots the distribution of the deviation of worker arrival times from the owner's preferred start time. $N = 60$ firms, 420 workers.

6 Figures



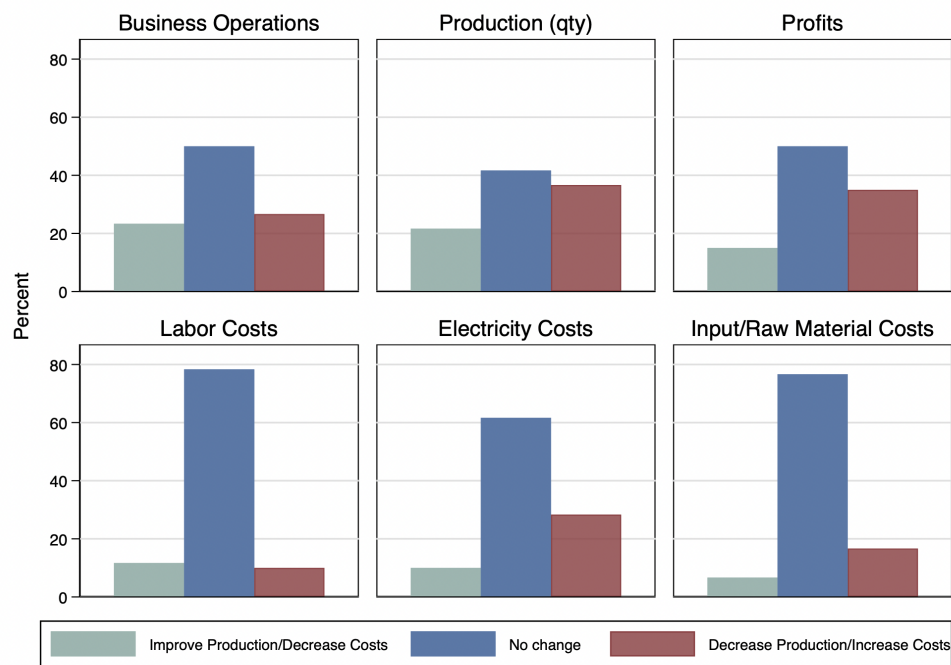
(a) Measures that could help increase production



(b) Why is coordination in worker arrival times important

Notes: Panel A reports owner's responses from the question "What type of measures would help increase production?", with options unprompted. $N = 60$. Panel B reports owner's responses from the question "If everybody starts work at the same time, how will it improve production?", with options 1 = Workers don't have to wait for others, 2 = Workers don't have to switch tasks, 3 = Workers are more productive during morning, and Others. This data is from the full sample of 60 firm owners.

Figure 3: How Providing Flexibility Affects Firm Operations



Notes: This figure reports owner's responses from the question: "Now I will list 4 different business objectives. For each of these, do you think providing workers flexibility affects this objective positively, negatively, or has no effect?" This data is from the full sample of 60 firm owners.

Figure 4: Experiment Design

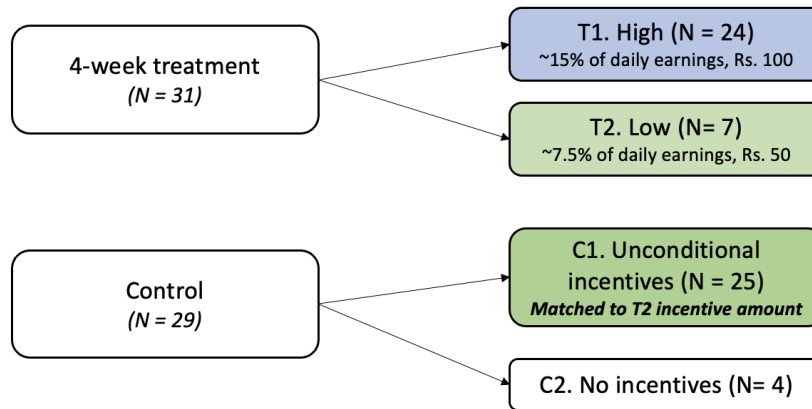


Figure 5: Study Timeline

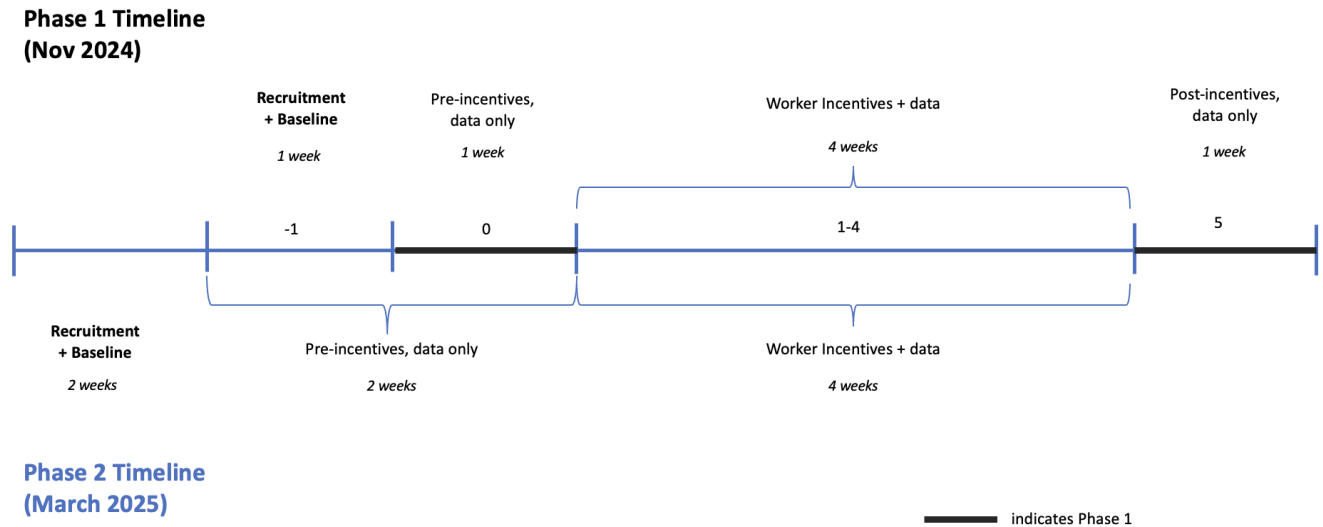
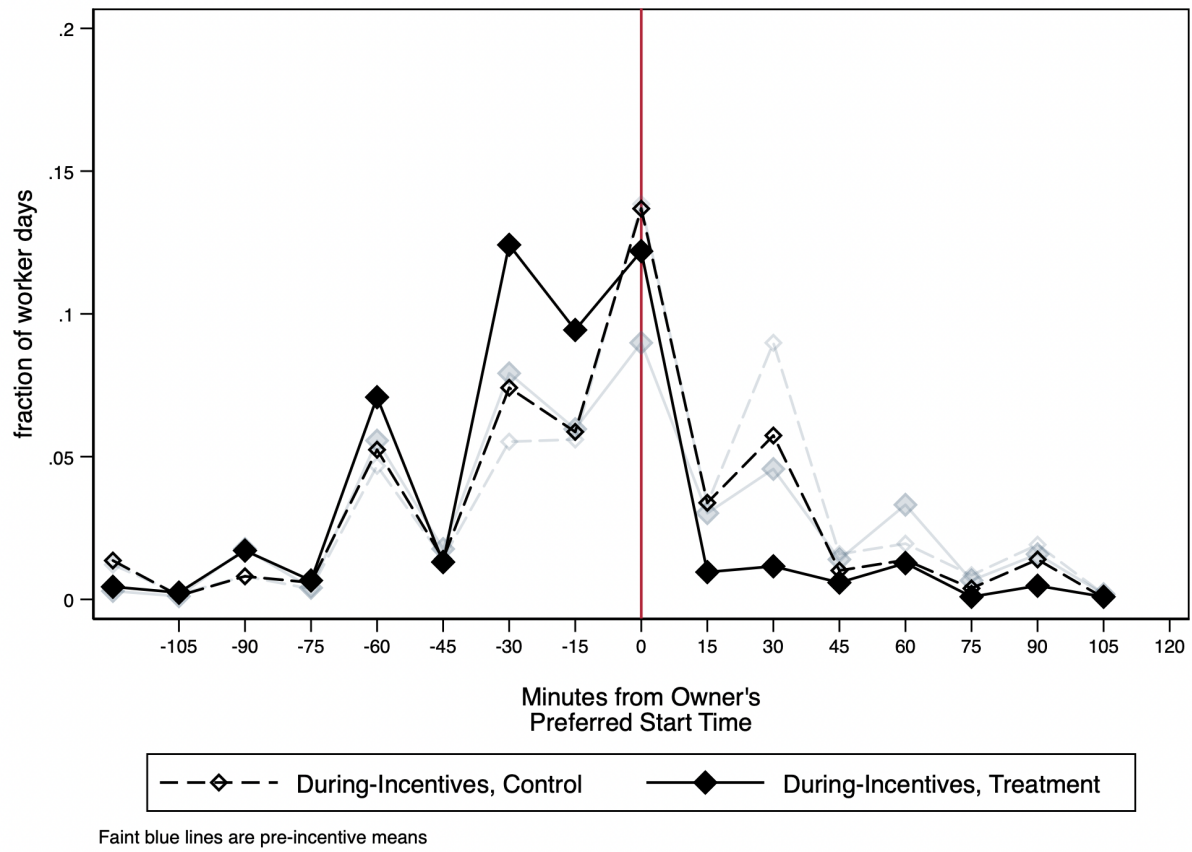


Figure 6: Change in deviation from owner's preferred start time



7 Tables

Table 1: Balance on Baseline Firm Characteristics

	Control	Treatment	T - C	<i>p-value</i>
	Mean/SD	Mean/SD	Difference	
Firm Characteristics				
# Tailors	6.28 (3.22)	5.97 (3.31)	-0.33	0.70
# Helpers	1.79 (1.15)	1.23 (1.06)	-0.58	0.03
Total # Machines	11.38 (4.52)	10.06 (3.03)	-1.35	0.17
Owner directly supervises workers	0.93 (0.26)	1.00 (0.00)	0.07	0.16
Firm Age (yrs)	7.38 (5.85)	5.30 (4.85)	-2.08	0.15
Produce own orders	0.28 (0.45)	0.03 (0.18)	-0.24	0.01
Export orders	0.14 (0.35)	0.06 (0.25)	-0.08	0.35
Weekly Output	4120.00 (2346.45)	4128.57 (3232.34)	96.74	0.89
Revenues (last 30 days)	147855.56 (137491.26)	119066.67 (104577.62)	-29888.47	0.28
Revenue per worker	19495.15 (14922.86)	17236.36 (13134.35)	-2360.59	0.49
Profits (last 30 days)	39459.56 (73012.58)	22633.33 (16643.28)	-17143.33	0.22
Profits per worker	4723.93 (5139.08)	3405.96 (2518.97)	-1342.23	0.20
Total Costs - wage bill (last 30 days)	31025.17 (21543.84)	32591.61 (21720.48)	1496.43	0.79
Owner Demographics				
Age	37.10 (7.87)	35.74 (6.97)	-1.31	0.50
Male	0.76 (0.44)	0.77 (0.43)	0.02	0.89
Completed Secondary School	0.52 (0.51)	0.39 (0.50)	-0.14	0.29
Commute Time (mins)	5.69 (6.33)	8.19 (9.62)	2.46	0.25
Experience in Industry (yrs)	18.83 (10.73)	16.70 (8.30)	-2.08	0.40
Owner does stitching work	0.90 (0.31)	0.81 (0.40)	-0.08	0.29
Owner stitches as a regular tailor	0.45 (0.51)	0.52 (0.51)	0.07	0.51
Observations	29	31		

Notes: This table reports means, mean differences by treatment group and p-values for baseline firm and owner characteristics. Mean differences and p-values are estimated using a regression of each variable on an indicator for treatment with phase x strata fixed effects, with robust standard errors.

Table 2: Balance on Baseline Worker Characteristics

	Control	Treatment	T - C	<i>p-value</i>
	Mean/SD	Mean/SD	Difference	
Age	36.62 (9.15)	36.48 (9.74)	-0.27	0.83
Male	0.44 (0.50)	0.41 (0.49)	-0.03	0.62
Married	0.93 (0.45)	0.91 (0.49)	-0.02	0.69
Has children < 14 yrs	0.86 (1.36)	0.84 (1.30)	-0.04	0.87
Has elderly member in HH	0.37 (0.48)	0.37 (0.49)	0.01	0.87
Related to owner	0.25 (0.43)	0.28 (0.45)	0.03	0.65
Native to TN	0.91 (0.29)	0.96 (0.20)	0.05	0.13
Commute < 15 mins	0.84 (0.37)	0.84 (0.36)	0.01	0.85
Completed Secondary School	0.46 (0.50)	0.40 (0.49)	-0.06	0.27
Experience in Industry (yrs)	14.70 (9.05)	14.34 (9.96)	-0.51	0.67
Tenure at Firm (yrs)	2.95 (4.03)	2.08 (3.07)	-0.88	0.16
Observations	189	187		

Notes: This table reports means, mean differences by treatment group and p-values for baseline firm and owner characteristics. Mean differences and p-values are estimated using a regression of each variable on an indicator for treatment with phase x strata fixed effects, with standard errors clustered at the firm level.

Table 3: Worker Attendance and Compliance

	Attendance	Compliance	Compliance cond. on Attendance
	(1)	(2)	(3)
Treatment $\times$ During Incentives	-0.00 [0.04]	0.11** [0.04]	0.16*** [0.04]
Control Mean (Pre-Incentives)	.75	.45	.6
Control Mean (Incentive Weeks)	.76	.53	.7
Treatment Effect %	0	.25	.27
Observations	10228	10228	7766

Notes: All specifications include worker, day-of-week, week and phase $\times$ strata fixed effects. Standard errors are clustered at the firm level. Attendance is defined as an indicator for whether a worker was present at the firm on that day. Compliance is defined as an indicator for whether a worker was present at the firm, *and* arrived by the owner's preferred start time on a particular day. The sample for these regressions excludes days on which a firm was shut, and no production took place.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 4: Heterogeneity in Treatment Effects on Compliance and Attendance

	Attendance		Compliance	
	(1)	(2)	(3)	(4)
Treatment $\times$ During	0.01 [0.04]	0.03 [0.05]	0.04 [0.05]	-0.02 [0.05]
Treatment $\times$ During $\times$ Female	-0.02 [0.04]		0.13** [0.05]	
Treatment $\times$ During $\times$ Late		-0.06 [0.06]		0.17*** [0.06]
Treatment $\times$ During $\times$ Very Late		-0.06 [0.08]		0.35*** [0.09]
Control Mean (Pre-Incentives)	.75	.75	.45	.45
Omitted Group Mean (Pre-Incentives)	.79	.68	.61	.63
Treatment Effect %	.02	.04	.09	-.04
<i>p-val</i> T $\times$ During $\times$ Female + T $\times$ During = 0	0.86		0.00	
<i>p-val</i> T $\times$ During $\times$ Late + T $\times$ During = 0		0.41		0.00
<i>p-val</i> T $\times$ During $\times$ Very Late + T $\times$ During = 0		0.69		0.00
Worker FE				
Observations	9905	10228	9905	10228

Notes: All specifications include worker, day-of-week, week, and phase $\times$ strata fixed effects, from the specification in [Equation 1](#). Standard errors are clustered at the firm level. Attendance is defined as an indicator for whether a worker was present at the firm on that day. Compliance is defined as an indicator for whether a worker was present at the firm, *and* arrived by the owner's preferred start time on a particular day. The sample for these regressions excludes days on which a firm was shut, and no production took place.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 5: Treatment Effects on Daily Work Hours

	Work Hours	Total # Breaks	Total Break Duration (mins)
	(1)	(2)	(3)
Treatment $\times$ During Incentives	0.03 [0.39]	0.16 [0.15]	-1.74 [3.28]
Control Mean Pre-Incentives	6.79	2.42	50.04
Control Mean Incentive Weeks	7	2.28	53.76
Treatment Effect %	0	.07	-.03
Observations	9934	5890	10228

Notes: All specifications include dow, week and phase $\times$ strata fixed effects. Standard errors are clustered at the firm level.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 6: Heterogeneity in Treatment Effects on Daily Work Hours

	Work Hours		Total # Breaks		Total Break Duration (mins)	
	(1)	(2)	(3)	(4)	(5)	(6)
Treatment $\times$ During	-0.07 [0.41]	0.09 [0.47]	0.14 [0.15]	0.08 [0.13]	-1.10 [3.32]	-1.34 [3.96]
Treatment $\times$ During $\times$ Female	0.22 [0.43]		0.03 [0.11]		-1.28 [3.49]	
Treatment $\times$ During $\times$ Late		-0.43 [0.55]		0.07 [0.15]		-0.87 [4.20]
Treatment $\times$ During $\times$ Very Late		0.25 [0.77]		0.24* [0.12]		-0.66 [5.95]
Control Mean (Pre-Incentives)	6.79	6.79	2.42	2.42	50.04	50.04
Omitted Group Mean (Pre-Incentives)	7.43	6.53	2.48	2.59	53.61	47.32
Treatment Effect %	-.01	.01	.06	.04	-.02	-.03
<i>p-val</i> T $\times$ During $\times$ Female + T $\times$ During = 0	0.74		0.31		0.57	
<i>p-val</i> T $\times$ During $\times$ Late + T $\times$ During = 0		0.43		0.47		0.55
<i>p-val</i> T $\times$ During $\times$ Very Late + T $\times$ During = 0		0.64		0.06		0.73
Worker FE	X	X	X	X	X	X
Observations	9616	9934	5788	5890	9905	10228

Notes: All specifications include day-of-week, week, phase $\times$ strata and worker fixed effects, from the specification in Equation 1. Standard errors are clustered at the firm level. Work hours are calculated as the difference between worker's exit and entry times, subtracting the total time spent on lunch, tea and other breaks during which the worker was not stitching. Total # of breaks includes the number of lunch, tea and other breaks taken by the worker during the day, and total break duration (in minutes) is the sum of time spent across all work breaks in a day.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 7: Normalized Worker Output and Earnings

Panel A: Production Quantity						
	Daily Output			Weekly Output		
	(1)	(2)	(3)	(4)	(5)	(6)
Treatment $\times$ During	0.08 [0.10]	0.05 [0.11]	0.05 [0.12]	0.27** [0.13]	0.26* [0.14]	0.15 [0.16]
Treatment $\times$ During $\times$ Female		0.06 [0.06]			0.07 [0.10]	
Treatment $\times$ During $\times$ Late			0.05 [0.09]			0.24 [0.16]
Treatment $\times$ During $\times$ Very Late			0.07 [0.12]			0.24 [0.21]
Control Mean (Pre-Incentives)	665.31	665.31	665.31	3667.43	3667.43	3667.43
Control SD (Pre-Incentives)	798.95	798.95	798.95	3965.94	3965.94	3965.94
Omitted Group Mean (Pre-Incentives)		718.07	668.32		4097.63	3643.73
Treatment Effect %	.09	.05	.06	.29	.28	.17
<i>p-val</i> T $\times$ During $\times$ Female + T $\times$ During = 0		0.33			0.02	
<i>p-val</i> T $\times$ During $\times$ Late + T $\times$ During = 0			0.41			0.01
<i>p-val</i> T $\times$ During $\times$ Very Late + T $\times$ During = 0			0.33			0.03
Worker FE	X	X	X	X	X	X
Observations	10729	9926	10729	1876	1742	1876
Panel B: Earnings						
	Daily Earnings			Weekly Earnings		
	(1)	(2)	(3)	(4)	(5)	(6)
Treatment $\times$ During	0.05 [0.09]	0.01 [0.11]	0.08 [0.10]	0.21* [0.12]	0.20 [0.15]	0.17 [0.14]
Treatment $\times$ During $\times$ Female		0.06 [0.08]			0.04 [0.12]	
Treatment $\times$ During $\times$ Late			-0.14 [0.11]			0.01 [0.18]
Treatment $\times$ During $\times$ Very Late			-0.01 [0.14]			0.13 [0.24]
Control Mean (Pre-Incentives)	587.81	587.81	587.81	3240.22	3240.22	3240.22
Control SD (Pre-Incentives)	708.81	708.81	708.81	3579.7	3579.7	3579.7
Omitted Group Mean (Pre-Incentives)		734.82	657.99		4193.17	3587.4
Treatment Effect %	.06	.01	.1	.23	.22	.19
<i>p-val</i> T $\times$ During $\times$ Female + T $\times$ During = 0		0.43			0.05	
<i>p-val</i> T $\times$ During $\times$ Late + T $\times$ During = 0			0.61			0.29
<i>p-val</i> T $\times$ During $\times$ Very Late + T $\times$ During = 0			0.60			0.17
Worker FE	X	X	X	X	X	X
Observations	10729	9926	10729	1876	1742	1876

Notes: SEs clustered at firm level. All specifications include day-of-week, week and phase $\times$ strata FE. This regression excludes owners/co-owners who also do stitching work at the firm. This regression excludes days of firm closure if no production took place at the firm. Output is normalized by product relative to pre-incentive week means. Workers who are absent on a given day are assumed to have worked on the product that most other workers in the firm worked on. Earnings are coded as 0 for workers who are absent on a given day. Output and earnings are winsorized at the 99th percentile before normalizing.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 8: Worker Output per Hour

	Daily Output per Hour			Weekly Output per Hour		
	(1)	(2)	(3)	(4)	(5)	(6)
Treatment $\times$ During	5.96 [7.39]	6.96 [8.24]	4.02 [8.28]	7.19 [5.86]	10.68 [7.04]	7.02 [7.18]
Treatment $\times$ During $\times$ Female		-2.75 [3.52]			-6.80 [4.46]	
Treatment $\times$ During $\times$ Late			10.02* [5.07]			5.48 [4.97]
Treatment $\times$ During $\times$ Very Late			-2.18 [5.91]			-4.29 [6.26]
Control Mean (Pre-Incentives)	99.34	99.34	99.34	95.03	95.03	95.03
Control SD (Pre-Incentives)	85.47	85.47	85.47	71.95	71.95	71.95
Omitted Group Mean (Pre-Incentives)		95.27	103.61		92.14	98.9
Treatment Effect %	.06	.07	.04	.08	.11	.07
<i>p-val</i> T $\times$ During $\times$ Female + T $\times$ During = 0		0.58			0.52	
<i>p-val</i> T $\times$ During $\times$ Late + T $\times$ During = 0			0.05			0.03
<i>p-val</i> T $\times$ During $\times$ Very Late + T $\times$ During = 0			0.82			0.67
Worker FE	X	X	X	X	X	X
Observations	7460	7346	7460	1508	1483	1508

Notes: SEs clustered at firm level. All specifications include day-of-week, week and phase $\times$ strata FE. This regression excludes owners/co-owners who also do stitching work at the firm. This regression excludes days of firm closure if no production took place at the firm. Output per hour is the total pieces worked on in a day across all products per hour (total hours in a day), and is winsorized at the 99th percentile.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 9: Treatment Effects on Weekly Firm Output - Pre vs. During Incentives x Treatment

	# Pieces		Revenues (INR)		Total Labor Cost (INR)	
	Production Data	Deliveries Data	Production Data	Deliveries Data	Production Data	Deliveries Data
	(1)	(2)	(3)	(4)	(5)	(6)
Treatment $\times$ During	0.31* [0.17]	0.64*** [0.18]	0.21 [0.12]	0.46*** [0.15]	0.16 [0.14]	0.84*** [0.31]
Control Mean (Pre-Incentives)	3200.55	3119.26	52402.26	57135.69	19080.13	19995.83
Control SD (Pre-Incentives)	3539.73	3576.75	76009	85156.04	25534.85	25060.2
Treatment Effect %	.34	.73	.3	.68	.21	1.05
p-value (Treatment $\times$ During)	0.08	0.00	0.10	0.00	0.28	0.01
Firm FE	X	X	X	X	X	X
Observations	118	118	118	118	118	118

Notes: All specifications include firm and phase $\times$ strata fixed effects. Standard errors are clustered at the firm level. Production/Delivery quantity is winsorized at the 99th percentile (by product, phase and incentive week). Revenues and total labor costs are calculated using winsorized quantities. All outcomes are normalized to pre-incentive, product level means to account for level differences in production quantities, and summed to the firm-week level, and then averaged to pre- and during incentive weeks.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

A Appendix Tables

Table A1: Worker Labor Supply and Production Outcomes - Pre Incentives

	All Workers	by Gender		by Avg. Arrival Time before incentives		
	(1)	Female (2)	Male (3)	Early (4)	0-30 mins late (5)	30+ mins late (6)
Demographics						
Female	0.58 (0.49)			0.39 (0.49)	0.68 (0.47)	0.80 (0.40)
Age	37.45 (9.57)	36.39 (7.92)	38.93 (11.30)	37.41 (10.38)	37.91 (8.53)	36.96 (9.21)
Married	0.82 (0.38)	0.91 (0.28)	0.79 (0.41)	0.74 (0.44)	0.87 (0.34)	0.92 (0.27)
# children < 14 yrs	0.89 (1.49)	1.08 (1.79)	0.62 (0.87)	0.65 (0.85)	0.97 (1.66)	1.23 (2.02)
Commute < 15 mins	0.83 (0.37)	0.87 (0.34)	0.79 (0.41)	0.82 (0.38)	0.85 (0.36)	0.84 (0.36)
Experience in Industry (yrs)	15.05 (9.67)	12.62 (8.49)	18.42 (10.18)	15.58 (9.99)	15.09 (9.32)	14.04 (9.42)
Attendance, Arrival Times						
Present	0.76 (0.42)	0.79 (0.41)	0.79 (0.41)	0.72 (0.45)	0.83 (0.37)	0.77 (0.42)
Minutes after owner's preferred start time	7.21 (50.32)	21.69 (47.29)	-11.72 (46.69)	-26.04 (38.32)	14.23 (35.18)	57.68 (39.87)
On-time or Early	0.47 (0.50)	0.30 (0.46)	0.65 (0.48)			
0-30 mins late	0.30 (0.46)	0.36 (0.48)	0.23 (0.42)			
30+ mins late	0.24 (0.42)	0.34 (0.47)	0.11 (0.32)			
Complied with Owner's Preferred Start Time	0.46 (0.50)	0.38 (0.48)	0.61 (0.49)	0.65 (0.48)	0.45 (0.50)	0.10 (0.30)
Total Work Hours	6.97 (4.24)	6.91 (4.01)	7.59 (4.25)	6.94 (4.62)	7.67 (3.76)	6.15 (3.84)
Total # Breaks	2.33 (0.93)	2.19 (0.88)	2.49 (0.97)	2.49 (0.94)	2.30 (0.85)	2.00 (0.93)
Total Break Duration - mins	51.72 (34.76)	51.57 (33.56)	55.42 (34.65)	49.98 (36.58)	56.66 (31.08)	48.91 (34.90)
Production						
# Unique Tasks	1.54 (1.44)	1.64 (1.43)	1.53 (1.45)	1.59 (1.62)	1.58 (1.33)	1.37 (1.14)
Non-tailoring work	0.12 (0.32)	0.12 (0.32)	0.13 (0.34)	0.11 (0.31)	0.16 (0.37)	0.08 (0.28)
Avg Task Piece Rate (INR)	0.86 (1.28)	0.77 (1.04)	1.06 (1.57)	0.97 (1.57)	0.84 (1.11)	0.68 (0.70)
Output per hours	101.93 (93.62)	99.13 (85.90)	104.68 (102.90)	111.27 (114.07)	97.42 (81.42)	90.63 (58.73)
Daily Earnings (INR)	548.17 (671.23)	491.99 (571.19)	668.15 (792.01)	618.13 (797.44)	559.00 (614.00)	396.68 (382.26)
Singer Task	0.17 (0.38)	0.14 (0.34)	0.24 (0.43)	0.20 (0.40)	0.17 (0.37)	0.13 (0.34)
Complex Singer Task	0.09 (0.28)	0.04 (0.21)	0.15 (0.36)	0.14 (0.34)	0.05 (0.23)	0.03 (0.16)
Observations	3020	1676	1210	1407	899	714

Notes: Standard deviations in parentheses. Data is at the worker-day level, including only 1-2 weeks of pre-incentives data from each phase. Work hours exclude the total duration of breaks taken during the day. Work hours, breaks and all production outcomes are 0 on days that the worker was absent.

Table A2: Treatment Effects on Deviation from Start Time

	Deviation from Start Time (mins)		
	(1)	(2)	(3)
Treatment $\times$ During	-8.13** [3.74]	-4.04 [4.50]	4.75 [3.74]
Treatment $\times$ During $\times$ Female		-7.58 [4.93]	
Treatment $\times$ During $\times$ Late			-14.82*** [4.62]
Treatment $\times$ During $\times$ Very Late			-36.64*** [7.14]
Control Mean (Pre-Incentives)	5.97	5.97	5.97
Omitted Group Mean (Pre-Incentives)		-13.6	-29.23
Treatment Effect %	-1.36	-.68	.8
<i>p-val</i> T $\times$ During $\times$ Female + T $\times$ During = 0		0.01	
<i>p-val</i> T $\times$ During $\times$ Late + T $\times$ During = 0			0.00
<i>p-val</i> T $\times$ During $\times$ Very Late + T $\times$ During = 0			0.00
Worker FE			
Observations	7778	7658	7778

Notes: All specifications include day-of-week, week, phase $\times$ strata and worker fixed effects, from the specification in Equation 1. Standard errors at clustered at the firm level. The sample for these regressions excludes days on which a firm was shut, and no production took place.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table A3: Treatment Effects on Coordination - Firm Level

	Mins between 1st and last worker (1)	Mins between the 10th and 90th pctl worker (2)	Deviation from start-time: 90th pctl worker (3)
Treatment $\times$ During Incentives	-19.89 [12.56]	-11.02 [7.49]	-13.62** [6.69]
Control Mean Pre-Incentives	130.71	85.82	49.33
Control Mean Incentive Weeks	107.84	67.44	28.11
Treatment Effect %	-.15	-.13	-.28
Observations	1590	1590	1590

Notes: This table reports firm-day level outcomes on coordination in worker arrival times. All specifications include dow, week and phase $\times$ strata fixed effects. Standard errors are clustered at the firm level. All outcome variables are in minutes, and are only computed for firm-days where strictly greater than 1 worker was present.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table A4: Worker Output and Earnings

Panel A: Production Quantity						
	Daily Output			Weekly Output		
	(1)	(2)	(3)	(4)	(5)	(6)
Treatment $\times$ During	37.55 [56.15]	27.59 [63.91]	25.42 [64.36]	546.70** [232.47]	532.11* [273.24]	351.03 [285.21]
Treatment $\times$ During $\times$ Female		15.15 [33.61]			41.62 [198.19]	
Treatment $\times$ During $\times$ Late			1.18 [57.31]			300.44 [317.53]
Treatment $\times$ During $\times$ Very Late			45.11 [64.04]			487.06 [408.87]
Control Mean (Pre-Incentives)	647.06	647.06	647.06	3233.28	3233.28	3233.28
Omitted Group Mean (Pre-Incentives)		704.48	644.83		3753.29	3224.15
Treatment Effect %	.06	.04	.04	.17	.16	.11
<i>p-val</i> T $\times$ During $\times$ Female + T $\times$ During = 0		0.49			0.04	
<i>p-val</i> T $\times$ During $\times$ Late + T $\times$ During = 0			0.69			0.03
<i>p-val</i> T $\times$ During $\times$ Very Late + T $\times$ During = 0			0.31			0.03
Worker FE	X	X	X	X	X	X
Product FE	X	X	X	X	X	X
Observations	11035	10221	11035	2295	2097	2295
Panel B: Earnings						
	Daily Earnings (INR)			Weekly Earnings (INR)		
	(1)	(2)	(3)	(4)	(5)	(6)
Treatment $\times$ During	40.46 [47.97]	32.40 [63.89]	65.06 [50.66]	401.43** [193.61]	393.63 [274.98]	434.75** [215.27]
Treatment $\times$ During $\times$ Female		16.31 [41.30]			48.00 [216.07]	
Treatment $\times$ During $\times$ Late			-91.09 [57.00]			-225.18 [278.11]
Treatment $\times$ During $\times$ Very Late			-16.96 [48.60]			68.92 [294.04]
Control Mean (Pre-Incentives)	571.68	571.68	571.68	2856.64	2856.64	2856.64
Omitted Group Mean (Pre-Incentives)		720.91	634.86		3840.8	3174.3
Treatment Effect %	.07	.06	.11	.14	.14	.15
<i>p-val</i> T $\times$ During $\times$ Female + T $\times$ During = 0		0.31			0.03	
<i>p-val</i> T $\times$ During $\times$ Late + T $\times$ During = 0			0.70			0.48
<i>p-val</i> T $\times$ During $\times$ Very Late + T $\times$ During = 0			0.40			0.10
Worker FE	X	X	X	X	X	X
Product FE	X	X	X	X	X	X
Observations	11035	10221	11035	2295	2097	2295

Notes: SEs clustered at firm level. All specifications include worker, product, week, and phase $\times$ strata fixed effects. This regression excludes owners/co-owners who also do stitching work at the firm. This regression excludes days of firm closure if no production took place at the firm. Output and earnings are coded as 0 for workers who are absent on a given day. All outcomes are winsorized at the 99th percentile.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table A5: Treatment Effects on Weekly Firm Output

	# Pieces		Revenues (INR)		Total Labor Cost (INR)	
	Production Data	Deliveries Data	Production Data	Deliveries Data	Production Data	Deliveries Data
	(1)	(2)	(3)	(4)	(5)	(6)
Treatment $\times$ During	0.30 [0.21]	0.65*** [0.22]	0.29 [0.20]	0.50*** [0.19]	0.18 [0.21]	0.61** [0.30]
Control Mean (Pre-Incentives)	3252.78	2941.86	48878.61	50293.51	19648.48	19493.93
Control SD (Pre-Incentives)	3986.25	3590.85	70087.69	79652.75	27196.3	28090.46
Treatment Effect %	.37	.8	.42	.79	.26	.88
p-value (Treatment $\times$ During)	0.15	0.00	0.16	0.01	0.39	0.05
Firm FE	X	X	X	X	X	X
Observations	336	336	336	336	336	336

Notes: All specifications include firm, week and phase $\times$ strata fixed effects. Standard errors are clustered at the firm level. Production/Delivery quantity is winsorized at the 99th percentile (by product, phase and incentive week). Revenues and total labor costs are calculated using winsorized quantities. All outcomes are normalized to pre-incentive, product level means to account for level differences in production quantities, and summed to the firm-week level.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table A6: Flexibility by Firm Size

	Small Firms		Large Firms	
	Mean	SD	Mean	SD
Timings Same	0.49	(0.50)	0.77	(0.42)
Timings Strict	0.15	(0.36)	0.50	(0.51)
Entry time is recorded	0.03	(0.17)	0.50	(0.51)
Pay cut if late	0.06	(0.23)	0.40	(0.49)
Margin of Lateness				
Acceptable to be 5 minutes late	0.97	(0.17)	0.94	(0.24)
Acceptable to be ≤ 30 minutes late	0.91	(0.29)	0.69	(0.47)
Acceptable to be > 30 minutes late	0.85	(0.36)	0.62	(0.49)
Observations	70		48	

Notes: SD in parenthesis (even columns). Small firms are defined as firms with up to 20 workers, and where the owner is often both a tailor (either full or part-time) and manager of the firm. Large firms include firms with at least 50 workers, where ownership and factory-floor management are separate.